Modular Forms

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Preface

Today, September 1st 2026, marks my 25th birthday. Usually, resolutions are made around New Year's, but to me this seemed like an appropriate time as well. I have self-studied (pure) mathematics on the side before, but never thoroughly nor was it well-structured. This time I intend to change that and aim to study the lecture notes on modular forms by Dr P. Bruin and Dr S. Dahmen entirely and complete all the problems. In this document I will write my solutions and include thoughts I have whilst reading the text.
I am in no rush, so an additional challenge is that I will not use any form of AI in solving the problems. The reason is simply that I notice that it becomes too easy and I believe in the value of the struggle of not knowing how to tackle a problem. Of course, I will allow myself to use the more 'classic' sources such as Wikipedia, StackExchange and papers. For this reason, my solutions might contain errors.[1] This project really is a challenge to me and my patience. Knowing myself, I would love to learn all sorts of topics in pure mathematics, so preferably as fast as possible too. To that end, I force myself to only focus on Modular Forms and not use AI to solve problems quicker. I believe (or hope) that the result will be more rewarding than superficially studying several subjects at once in a short time. After all, this project is my own so who am I kidding in that case?
Anyhow, I'm rambling a bit. Why modular forms then? The honest answer is no particular reason other than there being some inexplicable attraction. One quickly reads about the importance of the subject in maths and theoretical physics. I have yet to discover that, but for now I am satisfied with the curiosity it ignited in me.

Structure

For each chapter I will start by stating my thoughts and questions I had whilst reading the main text. I will indicate this by ***Pensées***. These comments might be silly and hopefully I will look back at them later and be able to answer them. When I (think I) know the answer, still, I will leave the questions here and edits by future me will be indicated in red.
I will attempt to solve all the problems. I would like to stress that my solutions will be far from polished and aim to show my thinking. Therefore, in the solutions, too, there will be modifications in red making them possibly lengthy.

With the disclaimers out of the way, let's get started.

[1] For someone reading this document and spotting errors, please let me know: kianshams2001@gmail.com.

1 The modular group

Pensées

- It is explained that 'modular' refers to the moduli space of complex elliptic curves, but what is 'form' actually? I know of differential forms, bi-linear forms etc. and then how are these modular forms related to said moduli space? Hopefully the relationship with elliptic curves will be clear when reaching the corresponding chapter.

- I'm curious why $\mathbb{R}$-linear independence is necessary in the definition of a lattice. Although they do not explicitly define this, I think $\omega_1, \omega_2 \in \mathbb{C}$ are $\mathbb{R}$-linearly dependent if $\exists \lambda \in \mathbb{R}$ such that $\omega_1 = \lambda\omega_2$ (or vice versa). I think this is the 'full rank' part of a lattice in $\mathbb{C}$ in the definition, because else the lattice would be one-dimensional. If so, making this explicit would have been nice. Also because any lattice $\omega_1\mathbb{Z} + \omega_2\mathbb{Z} = \omega_2\Lambda_z$ for $z \in \mathbb{C} \setminus \mathbb{R}$ by the assumption of $\mathbb{R}$-linear independence. Wlog $z = \omega_1/\omega_2 \in \mathbb{H}$ so any homogeneous function $F : \mathcal{L} \longrightarrow \mathbb{C}$ is completely determined by its values on Λ_z so induces a function $f : \mathbb{H} \longrightarrow \mathbb{C}$ via $f = F(\Lambda_z)$ for $z \in \mathbb{H}$ Also, why then are we only concerned with full rank lattices?

- Cool! Lattices in $\mathbb{C}$ are in 1-to-1 with complex elliptic curves. Curious to learn how that works later. (Actually, I also know already (merely as a trivia fact) that complex elliptic curves are Riemann surfaces. For the example of a torus I know how a lattice corresponds to it. Still, the general case will become clear later I hope.)

- They say that one is naturally led to study a function $F : \mathcal{L} \longrightarrow \mathbb{C}$ satisfying $F(\lambda\Lambda) = F(\Lambda)$, $\forall \lambda \in \mathbb{C}^*$ (this is modulo homothety), but turns out to be too restrictive. Hopefully, it will be clear why that is the case. Not explained at this point. But accepting this fact, homogeneous functions of weight k are the next best alternative (?) which motivates the study of modular forms later (I secretly already know a tiny bit about it, e.g. the definition). Example of homogeneous function of weight k is the Eisenstein series, which I think is also a modular form and is relevant in physics (read: string theory) too.

1.1

(i) " $\Longleftarrow$ ": This direction we assume $\begin{pmatrix}\omega_1' \\ \omega_2'\end{pmatrix} = \gamma \begin{pmatrix}\omega_1 \\ \omega_2\end{pmatrix}$ for some $\gamma = \begin{pmatrix}a & b \\ c & d\end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})$. Then $\omega_1' = a\omega_1 + b\omega_2$ and $\omega_2' = c\omega_1 + d\omega_2$, therefore $\Lambda' \subseteq \Lambda$.[2] Of course γ^{-1} exists so the converse also holds, i.e. $\Lambda' = \Lambda$.

" $\Longrightarrow$ ": $\Lambda' = \Lambda \implies \exists a, b, c, d \in \mathbb{Z}$ such that $\omega_1' = a\omega_1 + b\omega_2$ and $\omega_2' = c\omega_1 + d\omega_2$. Likewise $\exists \alpha, \beta, \delta, \theta \in \mathbb{Z}$ such that $\omega_1 = \alpha\omega_1' + \beta\omega_2'$ and $\omega_2 = \delta\omega_1' + \theta\omega_2'$. Plugging these relations in the first yields

$$\begin{aligned} & a(\alpha\omega_1' + \beta\omega_2') + b(\delta\omega_1' + \theta\omega_2') \overset{!}{=} \omega_1 \\ \Longrightarrow\ & a\beta + b\theta = 0 \ \wedge\ a\alpha + b\delta = 1, \quad \text{likewise} \\ & c(\alpha\omega_1' + \beta\omega_2') + d(\delta\omega_1' + \theta\omega_2') \overset{!}{=} \omega_2 \\ \Longrightarrow\ & c\alpha + d\delta = 0 \ \wedge\ c\beta + d\theta = 1. \end{aligned}$$

[2] Although the notation $\Lambda' \subseteq \Lambda$ is not introduced in the notes, I hope it is clear what I mean: all lattice points of Λ' are contained in Λ. This is different from homothety.

Compactly this is simply the requirement that

$$\underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\gamma} \begin{pmatrix} \alpha & \beta \\ \delta & \theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{1}$$

In other words $\gamma \in \mathrm{GL}_2(\mathbb{Z})$.

(ii) At first glance I don't understand what proposition 1.2 has to do with this part as stated in the hint... By elimination: part (i) and (ii) probably aren't useful...
Ah ok I see now, because for all $\gamma \in \mathrm{GL}_2(\mathbb{Z})$ $\det(\gamma) = \pm 1$. In order for $z' = \omega_1'/\omega_2'$ to remain in $\mathbb{H}$ we must have $\mathrm{Im}(z') > 0$. By proposition 1.2 (i) we have $\mathrm{Im}(\gamma z) = \frac{\det(\gamma)\mathrm{Im}(z)}{|j(\gamma,z)|^2} = \mathrm{Im}(z')$.
So we must have $\det(\gamma) > 0$, meaning $\det(\gamma) \overset{!}{=} 1$ or in other words $\gamma \in \mathrm{SL}_2(\mathbb{Z})$.
To make this proof complete really is just a copy of part (ii) and including this minor observation.

1.2

Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\gamma' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ be elements of $\mathrm{GL}_2(\mathbb{R})$. Then we simply work out

$$\begin{aligned} \gamma(\gamma' z) = \gamma\Big(\frac{a'z+b'}{c'z+d'}\Big) &= \frac{a\Big(\frac{a'z+b'}{c'z+d'}\Big) + b}{c\Big(\frac{a'z+b'}{c'z+d'}\Big) + d} \\ &= \frac{aa'z + ab' + bc'z + bd'}{ca'z + b'c + dc'z + dd'} \\ &= \frac{(aa'+bc')z + ab' + bd'}{(ca'+dc')z + b'c + dd'} =: (*). \end{aligned}$$

On the other hand,

$$\begin{aligned} (\gamma\gamma')\cdot z = \Big(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}\Big)\cdot z &= \begin{pmatrix} aa'+bc' & ab'+bd' \\ a'c+c'd & b'c+dd' \end{pmatrix}\cdot z \\ &= \frac{(aa'+bc')z + ab' + bd'}{(a'c+c'd)z + b'c + dd'} = (*). \text{ qed.} \end{aligned}$$

1.3

I am confused by the two parts of this question, because a *transitive action* of a group G on a set X means that for any $x, y \in X$ there exists a $g \in G$ such that $g \cdot x = y$. However, part b of this question then asks to show that for certain γ there are fixed points. How can the action be transitive and have a fixed point?
I am confusing two things here. If there is a γ for which there is an $x \in \mathbb{H}$ such that $\gamma \cdot x = x$ then there could still be a γ' such that $\gamma' \cdot x \neq x$ of course.

a. My initial idea was to brute force: compute $\gamma \cdot z = z'$ for $z, z' \in \mathbb{H}$ and show $\mathrm{Re}(\gamma \cdot z) = \mathrm{Re}(z')$, $\mathrm{Im}(\gamma \cdot z) = \mathrm{Im}(z')$ is always solvable, but this becomes messy.
There is a much cleaner idea which I owe to David C. Ullrich from the blogpost:

https://math.stackexchange.com/questions/2529664/how-to-show-the-action-of-sl-2-mathbb-r-on-the-complex-upper-half-plane-is-t . Namely, for any two points z, z' on the upper half plane first map e.g. point z' to i and then i to z. Because (**CLAIM**) we can map i to any point in the upper half plane by $\mathrm{SL}_2(\mathbb{R})$ and then we simply invert this and compose the two to find a map between the two points z, z'.

Let's prove **CLAIM**. First we need a map that sends $i \mapsto \lambda i$ for $\lambda \in \mathbb{R}_{>0}$ to fix the imaginary part and then $i \mapsto i + x$ for $x \in \mathbb{R}$ to fix the real part. The group elements that do this are $\begin{pmatrix} \sqrt{\lambda} & 0 \\ 0 & 1/\sqrt{\lambda} \end{pmatrix}$ and $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ which are both in $\mathrm{SL}_2(\mathbb{R})$. This proves $i \mapsto x + \lambda i$ upon composition.

Let's make it explicit that we can now use this to map any two points in the upper half plane with an $\mathrm{SL}_2(\mathbb{R})$ element:

Write $z' = \alpha + \beta i$, then $i \mapsto z'$ is implemented by

$$\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{\beta} & 0 \\ 0 & 1/\sqrt{\beta} \end{pmatrix} = \begin{pmatrix} \sqrt{\beta} & \alpha/\sqrt{\beta} \\ 0 & 1/\sqrt{\beta} \end{pmatrix} \tag{2}$$

so $z' \mapsto i$ is implemented by $\begin{pmatrix} \sqrt{\beta} & \alpha/\sqrt{\beta} \\ 0 & 1/\sqrt{\beta} \end{pmatrix}^{-1} = \begin{pmatrix} 1/\sqrt{\beta} & -\alpha/\sqrt{\beta} \\ 0 & \sqrt{\beta} \end{pmatrix}$. Likewise write $z = \epsilon + \delta i$ then $i \mapsto z$ via

$$\begin{pmatrix} \sqrt{\delta} & \epsilon/\sqrt{\delta} \\ 0 & 1/\sqrt{\delta} \end{pmatrix} \tag{3}$$

and to check we have

$$\begin{pmatrix} \sqrt{\delta} & \epsilon/\sqrt{\delta} \\ 0 & 1/\sqrt{\delta} \end{pmatrix} \begin{pmatrix} 1/\sqrt{\beta} & -\alpha/\sqrt{\beta} \\ 0 & \sqrt{\beta} \end{pmatrix} \cdot z' \tag{4}$$

$$= \begin{pmatrix} \sqrt{\frac{\delta}{\beta}} & -\alpha\sqrt{\frac{\delta}{\beta}} + \epsilon\sqrt{\frac{\beta}{\delta}} \\ 0 & \sqrt{\frac{\beta}{\delta}} \end{pmatrix} \cdot z' \tag{5}$$

$$= \frac{\sqrt{\frac{\delta}{\beta}} z' - \alpha\sqrt{\frac{\delta}{\beta}} + \epsilon\sqrt{\frac{\beta}{\delta}}}{\sqrt{\frac{\beta}{\delta}}} \tag{6}$$

$$= \frac{\delta}{\beta}(\alpha + \beta i) - \alpha\frac{\delta}{\beta} + \epsilon \tag{7}$$

$$= \epsilon + \delta i = z. \tag{8}$$

Where $\begin{pmatrix} 1/\sqrt{\beta} & -\alpha/\sqrt{\beta} \\ 0 & \sqrt{\beta} \end{pmatrix} \in \mathrm{SL}_2(\mathbb{R})$ proving transitivity of the action on $\mathbb{H}$.

b. I immediatly note that $a + d$ is the trace of the matrix, but what does this have to do with the fixed points? (nothing immediate I think)

Next, I thought $\gamma \cdot z \overset{!}{=} z$ does not yield a relation including an $a + d$ term, but $\gamma^2 \cdot z \overset{!}{=} \gamma \cdot z$ does. Then I collected terms and naively imposed the coefficient of z^2 terms to match, but this led to an inconsistency. That in turn made me realise we need to be careful because z^2 may not be in $\mathbb{H}$

eventhough z is (e.g. $z \in \mathbb{H}$ for $\mathrm{Re}(z) \in \mathbb{R}_{<0}$). So, I went back to $\gamma \cdot z \overset{!}{=} z$ to write

$$\begin{aligned} \gamma \cdot z = \frac{az+b}{cz+d} &\overset{!}{=} z \\ \iff cz^2 + (d-a)z - b &= 0 \end{aligned}$$

Recalling secondary-school mathematics, the roots are

$$z = \frac{-(d-a) \pm \sqrt{(d-a)^2 - 4c(-b)}}{2c} \tag{9}$$

and realising that $z \in \mathbb{H}$ we must have $(d-a)^2 - 4c(-b) < 0$ and only the $+$ in $\pm$ is in the upper half plane (corresponding to there only being one fixed point). Writing out the condition

$$\begin{aligned} &(d-a)^2 - 4c(-b) = d^2 - 2ad + a^2 + 4bc \\ &= d^2 + a^2 + 2bc - 2 \qquad (\text{using } ad - bc = 1) \\ &= (a+d)^2 - 2ad + 2bc - 2 \\ &= (a+d)^2 - 4 < 0 \iff |a+d| < 2. \end{aligned}$$

(N.B. This solution excludes $\gamma = \pm\mathbf{1}$, as stated in the problem, because then $c = 0$ which causes problems in equation 9.)

1.4

a. Write $H \coloneqq \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \,\middle|\, a, b \in \mathbb{R},\ a^2 + b^2 = 1 \right\}$. Then clearly $H \subseteq K$, because

$$\begin{aligned} \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \cdot i &= \frac{ai+b}{-bi+a} \\ &= \frac{(ai+b)(bi+a)}{a^2+b^2} \\ &= -ab + a^2 i + b^2 i + ab \\ &= (a^2+b^2)i = i. \end{aligned}$$

Conversely,

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot i = \frac{ai+b}{ci+d} &\overset{!}{=} i \\ \iff ai + b &= -c + di \\ \iff a = d \ \wedge\ b &= -c. \end{aligned}$$

Hence, $H = K$.

b. First I was categorically confused here because the quotient $\mathrm{SL}_2(\mathbb{R})/K$ is a group, so is $\mathbb{H}$ that too? Well, of course not with regular multiplication by the observation of my solution to 1.3b (z^2 need not be in $\mathbb{H}$ when z is). So it is purely set theoretical here.
Surjectivity follows from my solution 1.3a, because for any $i \neq z \in \mathbb{H}$ we can choose a $\gamma \in \mathrm{SL}_2(\mathbb{R})/K$ similar to the construction in equation 2. For $z = i$ we simply choose the identity of course.

Recall injectivity means $\phi(a) = \phi(b) \implies a = b$ for a map ϕ and a, b in the domain of ϕ. Well, then suppose $\gamma \cdot i = \gamma' \cdot i$. This means

$$\begin{aligned} \gamma'^{-1} \cdot (\gamma \cdot i) &= (\gamma'^{-1} \circ \gamma) \cdot i \qquad \text{(by prop 1.2.iii)} \\ &\overset{!}{=} i \end{aligned}$$

so in other words $\gamma'^{-1} \circ \gamma \in K$, hence $[\gamma] = [\gamma']$ in $\mathrm{SL}_2(\mathbb{R})/K$.

1.5

✓

2 Modular forms for $\mathrm{SL}_2(\mathbb{Z})$

Pensées

- Had to refresh this: 'A **meromorphic** function is a single-valued function that is analytic in all but possibly a discrete subset of its domain, and at those singularities it must go to infinity like a polynomial. A simpler definition states that a meromorphic function is a function $f(z)$ of the form $f(z) = \frac{g(z)}{h(z)}$ for g, h entire functions ($h(z) \neq 0$).' - from Wolfram Mathworld
- Then why do we require meromorphism in the definition of weakly modular?
- I understand the remark that for odd k we only have the zero function being weakly modular, but what is the relevance exactly (or why is that desired)?
- Awaiting my complex analysis self-study, I will for now accept the nice fact that the condition that $\tilde{f}$ can be continued to a meromorphic function on $\mathbb{D}$ is **equivalent** to stating

$$\tilde{f}(q) = \sum_{n=-\infty}^{\infty} a_n q^n, \qquad a_n \in \mathbb{C},\ a_n = 0 \text{ for } n \text{ sufficiently negative(?)} \tag{10}$$

  that is convergent on $\{q \in \mathbb{C} |\ 0 < |q| < \epsilon\}$ for some ϵ. (Whatever sufficiently negative means)
- Right now the definition of a modular form is quite abstract. I need some time to process all components of the definition and why they are relevant: holomorphicity, weakly modular (primarily what the factor of automorphy to the power of k means geometrically for a function).
- Ah I see why we need $k \geq 4$ for the Eisenstein series. First it needs to be even weight by the earlier observation and the proof shows that G_k only converges absolutely for $k \geq 3$.
- It is important that the Eisenstein series is induced by

$$\mathcal{G}_k : \mathcal{L} \longrightarrow \mathbb{C}, \tag{11}$$

$$\Lambda \mapsto \sum_{\omega \in \Lambda \setminus \{0\}} \frac{1}{\omega^k} \tag{12}$$

  with

$$G_k : \mathbb{H} \longrightarrow \mathbb{C}, \tag{13}$$

$$z \mapsto \mathcal{G}_k(\Lambda_z) \tag{14}$$

  because then it is a homogeneous function of weight k by proposition 1.4.
- The expansion of the sum above eq 2.3 is by simply observing

$$\sum_{(n,m) \in \mathbb{Z}^2 \setminus (0,0)} = \sum_{n \neq 0, m = 0} + \sum_{n = 0, m \neq 0} + \sum_{n,m \neq 0} \tag{15}$$

  and then collecting the last two sums as $\sum_{m \neq 0, n \in \mathbb{Z}}$
- A question came to mind: for g, h modular forms, is $f = g/h$ a modular form too? It satisfies the weakly modular part with weight of f being the difference in weights of g and h, but I think it fails the requirement of f being holomorphic on $\mathbb{H}$. Namely it is meromorphic because of potential zeros of h.

- Importance of lemma 2.5: summation order matters
- The first step in the proof of lemma 2.5:

$$\begin{aligned}\sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(mz+n)^2} &= \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(mz+n)^2} - \overbrace{\sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\Big(\frac{1}{mz+n}-\frac{1}{mz+n+1}\Big)}^{=0}\\ &= \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\Big(\frac{mz+n+1}{(mz+n)^2(mz+n+1)} - \frac{(mz+n)(mz+n+1)}{(mz+n)^2(mz+n+1)} + \frac{(mz+n)^2}{(mz+n)^2(mz+n+1)}\Big)\\ &= \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(mz+n)^2(mz+n+1)}\end{aligned}$$

- Adding some steps between the first and final expression of proof 2.4:

$$\begin{aligned}&2\zeta(2)z^{-2} + \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(nz-m)^2}\\ &= 2\sum_{j=1}^{\infty}\frac{1}{(jz)^2} + \sum_{n,m\in\mathbb{Z}}\frac{1}{(nz-m)^2} - \sum_{n\in\mathbb{Z}}\frac{1}{(nz)^2}\\ &= 2\sum_{j=1}^{\infty}\frac{1}{(jz)^2} + \sum_{n\neq 0}\sum_{m\in\mathbb{Z}}\frac{1}{(nz-m)^2} - \sum_{n\in\mathbb{Z}}\frac{1}{(nz)^2} + \sum_{m\in\mathbb{Z}}\frac{1}{m^2}\\ &= 2\zeta(2) + \sum_{n\neq 0}\sum_{m\in\mathbb{Z}}\frac{1}{(nz-m)^2}\end{aligned}$$

  Also, this fancy step is necessary

$$z^{-2}G_2(-1/z) + \frac{2\pi i}{z} = 2\zeta(2) + \sum_{n\in\mathbb{Z}}\sum_{m\neq 0}\frac{1}{(mz+n)^2(mz+n+1)}. \tag{16}$$

  because need to show absolute convergence in order to match the exchanged summation order in the definition of G_2. Now using this expression we can expand again as

$$\begin{aligned}&\sum_{n\in\mathbb{Z}}\sum_{m\neq 0}\frac{1}{(mz+n)^2(mz+n+1)} = \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(mz+n)^2(mz+n+1)}\\ &= \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\Big(\frac{1}{(mz+n)^2} + \frac{1}{mz+n} - \frac{1}{mz+n+1}\Big)\\ &= \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(mz+n)^2}\end{aligned}$$

- Initially I thought how on earth is the function Δ so strangely defined, but then I thought it is very reminiscent of the discriminant of an elliptic curve which I encountered in my thesis. Also the notation Δ is the same, so curious to see whether it indeed is that.
- I am not immediately familiar with the assertion that when the $\mathbb{C}$-vector space of weight 12 modular forms is one-dimensional and two forms have the same Fourier coefficient for q

that they must be equal. I get that the one-dimensionality implies they are scaled versions of each other, but why does one Fourier coefficient fix equality? Ah this is actually quite trivial. One-dimensionality means $\Delta = c\eta^{24}$ for some $c \in \mathbb{C}$. Then Fourier expansions are $\Delta = \sum_{n\geq 0} a_n q^n$, $\eta^{24} = \sum_{n\geq 0} b_n q^n$ so we get $a_n = cb_n$ and using $a_0 = 1 \overset{!}{=} cb_0 = c$ the equality follows.

- The proof of thm 2.9 is incomplete, eventhough it is quite trivial let me add: Using the valence formula and the fact that for holomorphic functions on $\mathbb{H} \cup \{\infty\}$ all orders are nonnegative integers, we have $k/12 = 1/3$ (for E_4) and $1/2$ (for E_6). The only way this is possible is if $\text{ord}_\rho E_4 = 1$ and $\text{ord}_i E_6 = 1$ and all other orders vanish.
- Corollary 2.10 implicitly answers an important fact: multiplication of two modular forms is another modular form (division is not, by my comment earlier) so does it form a ring? I think yes because $f = 1$ is a modular form which provides the identity and all other axioms seem satisfied. The function $0 : \mathbb{H} \longrightarrow \mathbb{C}$, $z \mapsto 0$ should be the additive identity. One subtlety is of course the weight.
- Loving the simple yet elegant proof of thm 2.12! Because indeed corollary 2.13 allows us to prove two modular forms are identical by comparing finitely many coefficients.

2.1

To be completely honest I did not know the definition of a group action. It does make intuitive sense (Wolfram MathWorld): A group action of a group G on a set X is a map $\phi : G \times X \longrightarrow X$ such that for all $x \in X$

1. $\phi(e, x) = x$ for e the identity in G
2. $\phi(g, \phi(h, x)) = \phi(gh, x)$ for all $g, h \in G$

So we simply check these two requirements.

1. $(f|_k \mathbf{1})(z) = (0z + 1)^{-k} f(\mathbf{1} \cdot z) = f(z) \longrightarrow$ ✓
2. $$\begin{aligned}(f|_k\gamma)(z) &= (cz + d)^{-k} f(\gamma \cdot z) =: g(z) \implies \\ (g|_k\gamma')(z) &= (c'z + d')^{-k} g(\gamma' \cdot z) \\ &= (c'z + d')^{-k}(c(\gamma' z) + d)^{-k} f(\gamma(\gamma' z)) \\ &= (c(a'z + b') + d(c'z + d'))^{-k} f((\gamma\gamma')z) = (*)\end{aligned}$$

   On the other hand, $\gamma\gamma' = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} aa' + bc' & ab' + bd' \\ a'c + c'd & b'c + dd' \end{pmatrix}$ so

   $$(f|_k(\gamma\gamma'))(z) = ((a'c + c'd)z + bc' + dd')^{-k} f((\gamma\gamma')z) = (*) \longrightarrow \checkmark$$

2.2

a. I mean this is quite by definition, because writing out $\sigma_t(m)\sigma_t(n)$ I get

$$\begin{aligned}&(1^t + d_{m_1}^t + d_{m_2}^t + \dots m^t)(1^t + d_{n_1}^t + d_{n_2}^t + \dots n^t)\\&= 1^{2t} + 1^t\sigma_t(n) + d_{m_1}^t\sigma_t(n) + \dots d_{m_i}^t\sigma_t(n) + \dots m^t\sigma_t(n) = \sigma_t(nm)\end{aligned}$$

where $d_{n_i}|n$, $d_{m_i}|m$. These are all the divisors of nm (raised to the power of t, as $1^{2t} = 1^t$ of course) since $d|n \implies d|nm$ and $d|m \implies d|nm$ so also $d_{n_i}d_{m_j}|nm$. In fact, all divisors of nm are of this form. Furthermore, because $\gcd(n, m) = 1$ there is also no over counting.

b.The desired expression screams geometric series: $\sum_{k=0}^{N} r^k = \frac{r^{N+1}-1}{r-1}$. Indeed, using the result of part (a.) we see inductively that

$$\sigma_t(n) = \prod_p \sigma_t(p^{e_p})$$

and

$$\begin{aligned}\sigma_t(p^{e_p}) &= \sum_{d|p^{e_p}} d^t = \sum_{j=0}^{e_p} p^{jt}\\&= \frac{p^{e_p+1} - 1}{p^t - 1}\end{aligned}$$

where I used the fact that p is prime, so all divisors of p^{e_p} are powers of p up to e_p. The desired expression follows.

2.3

a. Recall the generating function $\frac{t}{e^t-1} = \sum_{k=0}^{\infty} \frac{B_k}{k!}t^k$. Then we simply first send $t \mapsto 2\pi iz$:

$$\begin{aligned}\frac{t}{e^t - 1} = \frac{2\pi iz}{e^{2\pi iz} - 1} &= 2\pi iz\frac{1}{e^{\pi iz}(e^{\pi iz} - e^{-\pi iz})}\\&= \pi z\frac{1}{e^{\pi iz}\sin(\pi z)}\\&= \pi z\frac{e^{-\pi iz}}{\sin(\pi z)} = \sum_{k=0}^{\infty}\frac{B_k}{k!}(2\pi i)^k z^k.\end{aligned}$$

Likewise for $t \mapsto -2\pi iz$ I get

$$\begin{aligned}\frac{t}{e^t - 1} &= -2\pi iz\frac{1}{e^{-\pi iz}(e^{-\pi iz} - e^{\pi iz})}\\&= \pi z\frac{e^{\pi iz}}{\sin(\pi z)} = \sum_{k=0}^{\infty}\frac{B_k}{k!}(-2\pi i)^k z^k.\end{aligned}$$

Now if we add both terms I get

$$\begin{aligned}\pi z\frac{e^{\pi iz} + e^{-\pi iz}}{\sin(\pi z)} &= 2\sum_{k\geq 0 \text{ even}}\frac{B_k}{k!}(2\pi i)^k z^k\\\iff \pi z\frac{\cos(\pi z)}{\sin(\pi z)} &= \sum_{k\geq 0 \text{ even}}(2\pi i)^k\frac{B_k}{k!}z^k.\end{aligned}$$

b. Equation (A.1) reads $\pi\frac{\cos(\pi z)}{\sin(\pi z)} = \frac{1}{z} + \sum_{n=1}^{\infty}\Big(\frac{1}{z-n} + \frac{1}{z+n}\Big)$. In other words,

$$\begin{aligned}
&\frac{1}{z} + \sum_{n=1}^{\infty}\Big(\frac{1}{n(\frac{z}{n}-1)} + \frac{1}{n(\frac{z}{n}+1)}\Big) \\
&= \frac{1}{z} - \sum_{n=1}^{\infty}\sum_{k=0}^{\infty}\frac{1}{n}\Big((\frac{z}{n})^k - (-\frac{z}{n})^k\Big) \qquad \text{(geometric series)} \\
&= \frac{1}{z} - 2\sum_{\substack{k\geq 1\\ k \text{ odd}}}\sum_{n=1}^{\infty}\frac{1}{n}(\frac{z}{n})^k \\
&= \frac{1}{z} - 2\sum_{k\geq 2 \text{ even}} z^{k-1}\sum_{n=1}^{\infty}\frac{1}{n^k} \\
&= \frac{1}{z} - 2\sum_{k\geq 2 \text{ even}} \zeta(k) z^k
\end{aligned}$$

so multiplying both sides by z I indeed find

$$\frac{\pi z\cos\pi z}{\sin\pi z} = 1 - 2\sum_{\substack{k\geq 2\\ k \text{ even}}}\zeta(k)\, z^k. \tag{17}$$

c. This is straightforwardly writing out:

$$\begin{aligned}
\sum_{k\geq 0 \text{ even}}(2\pi i)^k\frac{B_k}{k!}z^k &= B_0 + \sum_{k\geq 2 \text{ even}}(2\pi i)^k\frac{B_k}{k!}z^k \\
&\overset{!}{=} 1 - 2\sum_{k\geq 2 \text{ even}}\zeta(k)z^k \qquad \forall z \implies \\
-2\zeta(k) &\overset{!}{=} (2\pi i)^k\frac{B_k}{k!} \iff \\
\zeta(k) &= -\frac{(2\pi i)^k}{2\cdot k!}B_k \qquad \forall k\geq 2
\end{aligned}$$

d. This requires looking at the generating function again:

$$\frac{t}{e^t-1} = \sum_{k=0}^{\infty}\frac{B_k}{k!}t^k.$$

I immediately thought we need to consider $t\mapsto -t$ in order to somehow get cancelling terms, as we want to show vanishing of B_k for $k>1$ odd. Upon close inspection I noticed something nice upon $t\mapsto -t$:

$$\begin{aligned}
&\frac{t}{e^t-1} \mapsto \frac{-t}{e^{-t}-1} = \frac{t}{1-e^{-t}} = \frac{te^t}{e^t-1} \implies \\
&\frac{te^t}{e^t-1} - \frac{t}{e^t-1} = t
\end{aligned}$$

Therefore, we must have

$$t \overset{!}{=} \sum_{k=0}^{\infty} \left(\frac{B_k}{k!}(-1)^k t^k - \frac{B_k}{k!} t^k \right)$$
$$\implies 1 \overset{!}{=} -2B_1 \quad \text{and} \quad B_k \overset{!}{=} 0 \, \forall k > 1 \text{ odd}.$$

2.4

a. Doesn't this immediately follow from its relationship with $E_4(z)$ which has a simple zero at $z = \rho = \exp\left(\frac{2\pi i}{3}\right)$ by thm 2.9? (yeah, but this question was probably intended to show a different way)
This I saw immediately geometrically by looking at the fundamental domain depicted in chapter 1.3. Namely, $\exp(2\pi i/3) = \rho$ so

$$\begin{aligned} G_4(-1/\rho) &= G_4(\rho + 1) \qquad \text{(by inspecting the figure (or simple algebra))} \\ &= G_4(\rho) \qquad \text{(modularity)} \\ &\overset{!}{=} \rho^4 G_4(\rho) \qquad \text{(by the hint provided)} \\ &= \rho G_4(\rho) \end{aligned}$$

so as $\rho \neq 0$, we must have $G_4(\rho) = 0$.

b. By the same procedure this one is a piece of cake. $G_6(-1/i) = G_6(i) \overset{!}{=} i^6 G_6(i) = -G_6(i) \implies G_6(i) = 0$.

2.5

I immediately note that $S \in \mathrm{SL}_2(\mathbb{Z})$ is satisfied by proposition 2.4 and captured by the transformation law written in the problem here. Also, T simply reads $E_2(z+1) = E_2(z)$ so that also checks out. The fact that $\mathrm{SL}_2(\mathbb{Z})$ is generated by S, T means that any element of the group is a product of powers of those matrices. Does it suffice to check ST and TS or something?
The ingredients are there, but I can't seem to cook the recipe yet. I will get back to this next week! (22-09-2026:) The idea I have is some kind of induction. Namely, every $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ is a product of powers of S, T. So given the proposed equation holds for S, T, then we can write $E_2(\gamma z) = E_2(Sw)$, with $w = S^{-1}\gamma z$, or $E_2(\gamma z) = E_2(Tv)$, with $v = T^{-1}\gamma z$ depending on whether there is a T or S on the left when γ is written as a product of them. Until eventually we simply end up with Sz or Tz in the argument. To that end I think it suffices to check 'associativity' of T^2, TS, ST and S^2 in the following sense.
Proposition 2.4 reads

$$E_2(S \cdot z) = z^2 E_2(z) - \frac{z}{4\pi i} \tag{18}$$

So we must have

$$\begin{aligned} E_2((S^2)z) &= E_2(Sw) \qquad (w = S\cdot z = -\frac{1}{z}) \\ &= w^2 E_2(w) - \frac{w}{4\pi i} \qquad \text{(by eq. 18)} \\ &= \frac{1}{z^2}E_2(-\frac{1}{z}) + \frac{1}{4\pi i z} \\ &= \frac{1}{z^2}\Big(z^2 E_2(z) - \frac{z}{4\pi i}\Big) + \frac{1}{4\pi i z} \qquad \text{(by eq. 18)} \\ &= E_2(z) \end{aligned}$$

And on the other hand $S^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ so the problem's proposed equation immediately reads $(0z-1)^{-2}E_2(S^2\cdot z)(= E_2(z)) \overbrace{=}^{\text{RHS}} E_2(z) - \frac{1}{4\pi i}\frac{0}{0z-1} = E_2(z)$ yielding the same result as above.

Likewise it is straightforward to show that TS, ST, T^2 also satisfy this associative property, but I remain unsure whether this is sufficient so I am a bit hesitant to write this out properly. Will get back to it later. I am also considering the option computing this associative property for $S^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $T^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with my proposed procedure. That at least 'feels' more like the right thing to do, although they are the same in spirit. The latter works as follows: $S^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -c & -d \\ a & b \end{pmatrix}$ and letting $w = S^{-1}\gamma z$, I get

$$\begin{aligned} E_2(\gamma z) &= E_2(Sw) \\ &= w^2 E_2(w) - \frac{w}{4\pi i} \qquad \text{(by prop. 2.4)} \\ &= \Big(\frac{-cz-d}{az+b}\Big)^2\Big((az+b)^2 E_2(z) - \frac{1}{4\pi i}\frac{a}{az+b}\Big) - -\frac{1}{4\pi i}\frac{cz+d}{az+b} \quad \text{(by induction on proposed formula)} \\ &= (cz+d)^2\Big(E_2(z) - \frac{1}{4\pi i}\frac{c}{cz+d}\Big) \qquad \text{(using } ad-bc=1) \end{aligned}$$

Likewise, $T^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a-c & b-d \\ c & d \end{pmatrix}$ so

$$\begin{aligned} E_2(\gamma z) = E_2(Tw') \qquad (w = T^{-1}\gamma z = \frac{(a-c)z+b-d}{cz+d}) \\ = E_2(w+1) = E_2(w) = (cz+d)^2(E_2(z) - \frac{1}{4\pi i}\frac{c}{cz+d}) \end{aligned}$$

so both agree with the proposed equation and like this we can strip away T or S iteratively from γ (which is a product of powers of T, S) until we are left with z in the argument of E_2.

2.6

a. This is relatively straightforward: $E_k(z) = \frac{(k-1)!}{2(2\pi i)^k} G_k(z)$ so $E_2(z) = \frac{1}{2(2\pi i)^2} G_2(z)$ thus

$$\begin{aligned}
f(\gamma z) &= G_2(\gamma z) - \frac{\pi}{\mathrm{Im}(\gamma z)} = 2(2\pi i)^2 E_2(\gamma z) - \frac{\pi}{\mathrm{Im}(\gamma z)} \\
&= 2(2\pi i)^2 \Big[j(\gamma, z)^2 \big(E_2(z) - \frac{1}{4\pi i} \frac{c}{cz+d} \big) \Big] - \frac{\pi}{\mathrm{Im}(\gamma z)} \qquad \text{(exercise 2.5)} \\
&= j(\gamma, z)^2 G_2(z) - j(\gamma, z)^2 2\pi i \frac{c}{cz+d} - \frac{\pi}{\mathrm{Im}(\gamma z)} \\
&= j(\gamma, z)^2 \Big[G_2(z) - 2\pi i \frac{c}{cz+d} \Big] - \frac{\pi}{\mathrm{Im}(z)} |cz+d|^2 \\
&= j(\gamma, z)^2 \Big[G_2(z) - 2\pi i \frac{c}{cz+d} - \frac{\pi(c\overline{z}+d)}{\mathrm{Im}(z)(cz+d)} \Big] \\
&= j(\gamma, z)^2 \Big[G_2(z) - \pi \frac{c(z-\overline{z})}{(cz+d)\mathrm{Im}(z)} - \frac{\pi(c\overline{z}+d)}{\mathrm{Im}(z)(cz+d)} \Big] \qquad (\mathrm{Im}(z) = \frac{1}{2i}(z-\overline{z})) \\
&= j(\gamma, z)^2 \Big[G_2(z) - \frac{\pi(cz+d)}{(cz+d)\mathrm{Im}(z)} \Big] = j(\gamma, z)^2 f(z).
\end{aligned}$$

b. Yes, $\frac{1}{\mathrm{Im}(z)}$ is holomorphic on $\mathbb{H} \cup \{\infty\}$ and part a. proves weight two weak modularity.

2.7

Recall $\Delta = \frac{(240E_4)^3 - (-504E_6)^2}{1728}$. Here I initially thought why is there a minus in front of 504, but it is to normalise the constant term to one:

$$E_4 = -\frac{B_4}{8} + \sum_{n=1}^{\infty} \sigma_3(n) q^n, \tag{19}$$

$$-\frac{B_4}{8} = --\frac{1}{8 \cdot 30} = \frac{1}{240} \tag{20}$$

$$E_6 = -\frac{B_6}{12} + \sum_{n=1}^{\infty} \sigma_5(n) q^n, \tag{21}$$

$$-\frac{B_6}{12} = -\frac{1}{504}. \tag{22}$$

My immediate idea was to expand the terms in Δ and seeing whether each coefficient is divisible by 1728.

$$\begin{aligned}
\Delta &= \frac{(1 + 240 \sum_{n=1}^{\infty} \sigma_3(n) q^n)^3 - (1 - 504 \sum_{n=1}^{\infty} \sigma_5(n) q^n)^2}{1728} \\
&= \frac{1}{1728} \Big[1 + 3 \cdot 240 \sum_{n=1}^{\infty} \sigma_3(n) q^n + 3 \big(240 \sum_{n=1}^{\infty} \sigma_3(n) q^n \big)^2 + \big(240 \sum_{n=1}^{\infty} \sigma_3(n) q^n \big)^3 \\
&- \Big(1 - 2 \cdot 504 \sum_{n=1}^{\infty} \sigma_5(n) q^n + \big(504 \sum_{n=1}^{\infty} \sigma_5(n) q^n \big)^2 \Big) \Big]
\end{aligned}$$

the 1's cancel and every coefficient is divisible by 1728 except

$$\sum_{n=1}^{\infty} \big(3 \cdot 240 \sigma_3(n) + 2 \cdot 504 \sigma_5(n) \big) q^n \tag{23}$$

which must therefore somehow be divisible by 1728 too... Ah, perhaps ideas from exercise 8,9 might be useful? Hmm, cannot have a relation with σ_3 and σ_5 because we would need $E_6 = cE_4E_2$, but E_2 is not a modular form and this would involve $\sigma_1(n)$ too.
Let's try something else. $1728 = 2^6 \cdot 3^3$, $3 \cdot 240 = 2^4 \cdot 3^2 \cdot 5$ and $2 \cdot 504 = 2^4 \cdot 3^2 \cdot 7$ so dividing out common terms,

$$\frac{1}{1728}\Big(\sum_{n=1}^{\infty}\big(3\cdot 240\sigma_3(n) + 2\cdot 504\sigma_5(n)\big)q^n\Big) = \frac{1}{12}\Big(\sum_{n=1}^{\infty}\big(5\sigma_3(n)+7\sigma_5(n)\big)q^n\Big)$$
$$\implies \text{must prove: } \frac{5\sigma_3(n)+7\sigma_5(n)}{12} \in \mathbb{Z}\ \forall n$$

Let's first see if this is true for prime numbers $n = p$. Then the only divisors are, by definition, 1 and p so $\sigma_3(p) = 1 + p^3$ and $\sigma_5(p) = 1 + p^5$ yielding

$$\frac{5p^3+7p^5}{12} = \frac{(5+7p^2)p^3}{12}, \checkmark \text{ for } p = 2, 3$$

Next for any $p > 3$ we must apparently have that $12|(5+7p^2)$ (as $12 = 2^2 \cdot 3$). To prove this, a fact I encountered during a group theory course came to min: $a|x$ and $b|x \implies \text{lcm}(a,b)|x$ so if we prove $3|(5+7p^2)$ and $4|(5+7p^2)$ we are done. Well

$$5 + 7p^2 \bmod 3 = \overline{2} + \overline{p^2} \stackrel{!}{=} \overline{0}$$
$$\implies \overline{p^2} = \overline{p}^2 \stackrel{!}{=} \overline{1}$$
$$\iff \overline{p} = \overline{1} \text{ or } \overline{p} = \overline{2}$$

which is true for any prime number $p > 3$. Likewise

$$5 + 7p^2 \bmod 4 = \overline{1} + \overline{3}\overline{p^2} \stackrel{!}{=} \overline{0}$$
$$\implies \overline{p^2} = \overline{p}^2 \stackrel{!}{=} \overline{1}$$
$$\iff \overline{p} = \overline{1} \text{ or } \overline{p} = \overline{3}$$

which is again true for any prime number $p > 3$ so $12|(5+7p^2)\ \forall p > 3$, thus $\frac{(5+7p^2)p^3}{12} \in \mathbb{Z}$ for every prime number.
Now we can probably use the result of problem 2.2 somehow...The generalisation is not obvious to me at the moment. Will get back to it after continuing with 11-15

It turns out that this can be proved quite elegantly, which I will explain now and owe to a colleague Manuel Van (MSc mathematics). We have established that this problem can be reduced to showing $12|(5\sigma_3(n) + 7\sigma_5(n))$. This in turn means

$$5\sigma_3(n) + 7\sigma_5(n) \equiv 0 \bmod 12 \iff$$
$$5\sigma_3(n) \equiv -7\sigma_5(n) \bmod 12 \equiv 5\sigma_5(n) \bmod 12 \iff$$
$$\sigma_3(n) \equiv \sigma_5(n) \bmod 12$$

which for $n = 1, 2$ is true and for $n = p$ prime ≥ 3 I want to give a more elegant argument than before. Namely $\sigma_t(p) = 1 + p^t$, so $\sigma_3(p) - \sigma_5(p) \equiv p^3 - p^5 = -p^3(p-1)(p+1) \equiv 0 \bmod 12$, because

$p-1, p, p+1$ are three consecutive numbers so it must contain a multiple of 3 and $p-1, p+1$ are even so in total it should be divisible by 12. Now the rest of the argument is straightforward because we know $\sigma_t(nm) = \sigma_t(n)\sigma_t(m)$ for $\gcd(n,m) = 1$ and we know now $\sigma_3(p) \equiv \sigma_5(p)$ for all p prime and $p = 1$, so for any positive integer $n = p_1^{e_1} \cdot \cdots \cdot p_r^{e_r}$ we obtain

$$\begin{aligned}
\sigma_3(n) &= \sigma_3(p_1^{e_1} \cdot \ldots \cdot p_r^{e_r}) \\
&= \sigma_3(p_1)^{e_1} \cdot \ldots \cdot \sigma_3(p_r)^{e_r} \\
&\equiv \sigma_5(p_1)^{e_1} \cdot \ldots \cdot \sigma_5(p_r)^{e_r} \bmod 12 \\
&= \sigma_5(p_1^{e_1} \cdot \ldots \cdot p_r^{e_r}) \bmod 12 \\
&= \sigma_5(n) \bmod 12.
\end{aligned}$$

2.8

$|M_8| = 1 \implies f = cE_8 = c\Big(-\frac{B_8}{16} + \sum_{n=1}^{\infty} \sigma_7(n)q^n\Big)$ for some $c \in \mathbb{C}$ and $f \in M_8$. We also know $E_4^2 \in M_8$ where $E_4 = \frac{1}{240} + \sum_{n=1}^{\infty} \sigma_3(n)q^n$ so

$$\begin{aligned}
E_4^2 &= \big(\frac{1}{240}\big)^2 + \frac{1}{120}\sum_{n=1}^{\infty}\sigma_3(n)q^n + \Big(\sum_{n=1}^{\infty}\sigma_3(n)q^n\Big)^2 \\
&\overset{!}{=} cE_8 \overbrace{\implies}^{\text{Cor. 2.13}} \frac{c}{480} \overset{!}{=} \frac{1}{240^2} \iff c = \frac{1}{120}.
\end{aligned}$$

In other words,

$$\frac{1}{120}\sum_{n=1}^{\infty}\sigma_7(n)q^n = \frac{1}{120}\sum_{n=1}^{\infty}\sigma_3(n)q^n + \Big(\sum_{n=1}^{\infty}\sigma_3(n)q^n\Big)^2 \tag{24}$$

and requiring this to hold for all q we find

$$\sigma_7(n) = \sigma_3(n) + 120\sum_{j=1}^{n-1}\sigma_3(j)\sigma_3(n-j). \tag{25}$$

2.9

a. Like the previous problem, $|M_{10}| = 1$ so $E_{10} = cE_4E_6$ for some $c \in \mathbb{C}$ yielding $-\frac{B_{10}}{20} \overset{!}{=} c\frac{-B_6}{12} \cdot \frac{-B_4}{8}$ by corollary 2.13 again $\implies c = \frac{5040}{11}$.

$$\begin{aligned}
E_{10} = cE_4E_6 = c\Big[\big(\frac{1}{240} + \sum_{n=1}^{\infty}\sigma_3(n)q^n\big)\big(-\frac{1}{504} + \sum_{n=1}^{\infty}\sigma_5(n)q^n\big)\Big] \implies \\
\sum_{n=1}^{\infty}\sigma_9(n)q^n = c\frac{1}{240}\sum_{n=1}^{\infty}\sigma_5(n)q^n - c\frac{1}{504}\sum_{n=1}^{\infty}\sigma_3(n)q^n + c\sum_{n=1}^{\infty}\sum_{j=1}^{n-1}\sigma_3(j)\sigma_5(n-j)q^n \implies \\
\sigma_9(n) = \frac{21}{11}\sigma_5(n) - \frac{10}{11}\sigma_3(n) + \frac{5040}{11}\sum_{j=1}^{n-1}\sigma_3(j)\sigma_5(n-j).
\end{aligned}$$

b. This follows the same line: $E_{14} = cE_4E_{10} \implies -\frac{1}{24} = c\frac{1}{240} \cdot \frac{-5}{20 \cdot 66} \implies c = 2640$ obtaining the relation

$$\sigma_{13}(n) = c\frac{-B_4}{8}\sigma_9(n) + c\frac{B_{10}}{20}\sigma_3(n) + c\sum_{j=1}^{n-1}\sigma_3(j)\sigma_9(n-j) \tag{26}$$

Likewise $E_{14} = \tilde{c}E_6E_8 \implies \tilde{c} = 10080$ yielding

$$\sigma_{13}(n) = \tilde{c}\frac{-B_6}{12}\sigma_7(n) + \tilde{c}\frac{-B_8}{16}\sigma_5(n) + \tilde{c}\sum_{j=1}^{n-1}\sigma_5(j)\sigma_7(n-j) \tag{27}$$

c. Really same Spiel.... $E_{14} = cE_4^2E_6$ finding $c = \frac{12B_{14}}{28B_6}\left(\frac{8}{B_4}\right)^2$ etc. etc., tedious but straightforward.

2.10

a. I think this statement is trivial by using thm 1.5: that every point in $\mathbb{H}$ is $\mathrm{SL}_2(\mathbb{Z})$ equivalent to a point in the fundamental domain. Because the lowest (right) corner is at $\rho + 1 = \frac{1+i\sqrt{3}}{2}$ which has imaginary part $C = \sqrt{3}/2$, qed.

b. A weight zero modular form f satisfies $f(\gamma z) = f(z)$, $\forall z \in \mathbb{H}$ and $\forall \gamma \in \mathrm{SL}_2(\mathbb{Z})$. By part a. we can thus assume $\mathrm{Im}(z) \geq \sqrt{3}/2$ and a cusp form f has q-expansion $f = \sum_{n=1}^{\infty} a_n q^n$ so

$$\begin{aligned} |f| &= \sum_{n=1}^{\infty} |a_n||q^n|, \\ |q^n| &= |\exp(2\pi i(\mathrm{Re}(z) + i\mathrm{Im}(z)))n| \\ &= \exp(-2\pi n\mathrm{Im}(z)) \leq \exp\Big(-\sqrt{3}\pi n\Big) \end{aligned}$$

so indeed $|f|$ attains a maximum on $\mathbb{H}$, precisely where the bound is saturated.

c. By thm A.4 the cusp form f must be constant on $\mathbb{H}$. I don't immediately see how I must now conclude that every cusp form must be zero and what remains is constant modular forms of weight zero... I guess that we can write for every $\tilde{f} \in M_0$ that $\tilde{f} = c + f$, for $c \in \mathbb{C}$ and $f \in S_0$ and I think the weight zero modularity property together with the fact that f is constant ought to force $f = 0$... Ah yes of course, because cusp forms satisfy $f(\infty) = 0$ which is only possible when $f = 0$ altogether (as $f =$const. by b.).

2.11

a. $\Delta = \lambda E_4^3 + \mu E_{12}$ can be solved for λ and μ using corollary 2.13 and the first few terms of the expansion of Δ; $a_0 = 0$, $a_1 = 1$. This quickly gives the system of equations

$$\begin{aligned} \lambda\big(-\frac{B_4}{8}\big) - \mu\frac{B_1 2}{24} &= 0, \\ 3\lambda\big(-\frac{B_4}{8}\big)^2\sigma_3(1) + \mu\sigma_{11}(1) &= 1. \end{aligned}$$

Noting that $\sigma_t(1) = 1 \forall t$ and $B_4 = -\frac{1}{30}$, $B_{12} = -\frac{691}{2730}$ we can let Mathematica solve this and I found $\lambda = 22112$, $\mu = -\frac{91}{600}$.

b. As $691|22112$, it immediately follows that $\Delta \equiv \mu E_{12} \bmod 691$. I.e. $600\Delta \equiv -91E_{12} \bmod 691 \equiv 600E_{12} \bmod 691$, hence $\tau(n) \equiv \sigma_{11}(n) \bmod 691$.

2.12

a. Let us write $g(z) = f'(z) + 4\pi i k E_2(z) f(z)$. We need to show $g \in M_{k+2}$ when $f \in M_k$. To that end I will first show weak modularity.

$$\begin{aligned} g(\gamma z) &= f'(\gamma z) + 4\pi i k E_2(\gamma z) f(\gamma z) \\ &= f'(\gamma z) + 4\pi i k (cz+d)^2 \Big(E_2(z) - \frac{1}{4\pi i} \frac{c}{cz+d} \Big) (cz+d)^k f(z) \quad \text{(by prop. 2.4 and } f \in M_k) \\ &= f'(\gamma z) + (cz+d)^{k+2} 4\pi i k E_2(z) f(z) - (cz+d)^{k+1} ck f(z). \end{aligned}$$

So if we can show $f'(\gamma z) = (cz+d)^{k+2} f'(z) + (cz+d)^{k+1} ckf(z)$ this part is done. We need to be careful in notation here: $f'(w)$ means $\frac{\partial f}{\partial w}$ and $(f(w)' = \frac{\partial f}{\partial w} \frac{\mathrm{d}w}{\mathrm{d}z}$

$$\Big(f(\gamma z) \Big)' = f'(\gamma z)(\gamma z)' = \frac{1}{(cz+d)^2} f'(\gamma z)$$

but also

$$\begin{aligned} \Big(f(\gamma z) \Big)' &= \Big((cz+d)^k f(z) \Big)' \\ &= ck(cz+d)^{k-1} f(z) + (cz+d)^k f'(z) \implies \\ f'(\gamma z) &= ck(cz+d)^{k+1} f(z) + (cz+d)^{k+2} f'(z), \ qed. \end{aligned}$$

!It remains to argue that $g(z)$ is holomorphic on $\mathbb{H} \cup \{\infty\}$!. Of course $f'(z)$ is holomorphic, because by assumption $f(z)$ is too.

b. I think we can show closure upon differentiation for monomials E_2, E_4, E_6 and qed, because then it follows for any polynomial by the product rule. In the previous part we found $g(z) = f'(z) + 4\pi i k E_2(z) f(z)$ and that for any $f \in M_k$ we have $g \in M_{k+2}$. Letting $f = E_2$, I obtain $g = E_2' + 8\pi i E_2^2$, but also $\dim(M_4) = 1$ so $g = \alpha E_4$ for some $\alpha \in \mathbb{C}$. Hence, $E_2' = \alpha E_4 - 8\pi i E_2^2 \in \mathbb{C}[E_2, E_4, E_6]$. We play the same game for E_4, E_6: $f = E_4 \implies E_4' = \alpha E_6 - 16\pi i E_2 E_4$ (not necessarily the same α as before) and for $f = E_6 \implies E_6' = \alpha E_8 - 24\pi i E_2 E_6$, but $E_8 = \beta E_4^2$ for some $\beta \in \mathbb{C}$ as $\dim(M_8) = 1$ as well. I think this is the idea, but it feels a bit shaky still as E_2 is not a modular form. However, the transformation rule of part a. still holds for E_2 too.

2.13

a. This seems rather obvious, because modular functions are of the form f/g for $f, g \in M_k$ and $g \neq 0$. Then $1 \in F$ (take $f = 1 = g$) and the inverse is always given (for non-zero elements) by g/h.

b. Do they mean the j-function here?? I'm super confused because I am only familiar with 'numbers' being transcendental over some field (like π over $\mathbb{Q}$) and since $\mathbb{C}$ is algebraically closed I'm a bit puzzled. But then I looked it up and functions can also be transcendental by the same requirement: If $\nexists f \in \mathbb{C}[x]$ such that $f(j) = 0$ (the zero function), then j is a transcendental function over $\mathbb{C}$. But I know that for $K \subseteq \overline{K} \subseteq L$, with $\overline{K}$ the algebraic closure of K, we have that for all elements $\alpha \in L$ not contained in $\overline{K}$ that α is transcendental over K. Does it then suffice to remark that $j \notin \mathbb{C}$?? Definitely not, because then you could say that any polynomial is a transcendental function... I need to figure out how this translates to transcendental functions.

2.14

a. We know $j = \frac{(240E_4)^3}{\Delta}$, with $\Delta = \frac{(240E_4)^3 - (-504E_6)^2}{1728}$, and $E_k \propto G_k$. In problem 4 we saw that $G_4(\rho) = 0$ and $G_6(i) = 0$, so $E_4(\rho) = 0 = E_6(i)$. Using this, it immediately follows that $j(i) = 1728$ and $j(\rho) = 0$.

b. $\mathrm{Re}(z) = 0$ is easy, because then $E_4(z)$ and $E_6(z)$ are real and therefore $j(z) \in \mathbb{R}$.[3] Also the left and right boundary edges of $\mathcal{D}$ are quite straightforward because $q(\mathrm{Re}(z) = \pm\frac{1}{2}) = \exp\big(2\pi i(\pm\frac{1}{2} + bi)\big) = \exp(\pm\pi i)\exp(-2\pi b) \in \mathbb{R}$ (so the entire q-expansion of j is real too).
It remains to show the arc part of the boundary of $\mathcal{D}$. To that end we can exploit $j(\gamma z) = j(z)$ for all $j \in \mathrm{SL}_2(\mathbb{Z})$. Namely on this part of the boundary $|z| = 1$ so given $z \sim -\frac{1}{z} = -\frac{\overline{z}}{|z|^2} = -\overline{z}$ we must have

$$\begin{aligned} j(z) &= j(-\overline{z}) \\ &= \overline{j(z)} \implies j(z) \in \mathbb{R} \end{aligned}$$

where the last step can be seen by $\overline{q} = \exp(-2\pi i\overline{z}) = q(-\overline{z})$.

c. Well-definedness follows immediately by $j(\gamma z) = j(z)\ \forall\gamma \in \mathrm{SL}_2(\mathbb{Z})$. The bijective part of this question has been bugging me for days... (!)

d. Requiring $j(z) = \overline{j(z)} = j(-\overline{z}) \implies$ there must be a $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that $z = \gamma \cdot -\overline{z}$. Since we take all $z \in \mathcal{D}$ (representative of domain of the map $\overline{j}$), we have 4 cases.

1. $\gamma = S$: $z = \frac{1}{\overline{z}} \implies |z| = 1$ so this is the circular arc of $\mathcal{D}$.
2. $\gamma = T$: $z = -\overline{z} + 1 \iff a + bi = 1 - a + bi \implies a = \frac{1}{2}$. So this is the right boundary of $\mathcal{D}$.
3. $\gamma = T^{-1}$: $z = -\overline{z} - 1 \iff a + bi = -1 - a + bi \implies a = -\frac{1}{2}$. So this is the left boundary of $\mathcal{D}$.
4. $\gamma = \mathbf{1}$: $z = -\overline{z}$ is satisfied when $\mathrm{Re}(z) = 0$.

2.15

a. First I was confused 'what about M_2, M_0?', but $|M_2| = 0$ of course (and $4a + 6b = 0$ has no solutions) and $|M_0| = 1$ (corresponding with the unique solution $(a, b) = (0, 0)$ of $4a + 6b = 0$). I secretly already use the result of part b that E_4 and E_6 are independent so the different solutions to the equation correspond with the dimension of M_k. Next, I thought what's special about 2? But then I saw that $k = 0, 4, 6, 8, 10$ all have a single solution for $4a + 6b = k$ and $k = 12$ has two, but $k = 2$ has none and I realised that for every multiple of 12 you get an extra solution to the equation. That is because suppose $4a+6b = x$ is a solution, then $4a'+6b' = x+12$ has an additional solution: $(a', b') = (a + 3, b)$ or $(a', b') = (a, b + 2)$. The solution to the first equation gets shifted to a new solution and there is an additional one. Of course this would mean that every solution would provide a new solution when $k \mapsto k + 12$, but this would overcount in for example $4a + 6b = 12$: it has solutions $(3, 0)$, $(0, 2)$ and $4a' + 6b' = 24$ would then give $(6, 0)$, $(3, 2)$ and $(3, 2)$, $(0, 4)$ as new solutions which overcounts the middle solution $(3, 2)$.

b. I had to look up algebraic independence from Wikipedia: A subset $S \subseteq L$ of a field L

[3] This can be seen by the q-expansion and recalling $q = \exp(2\pi iz)$.

is algebraically independent over $K \subseteq L$ if there exists no non-zero polynomial $P(x_1, \ldots, x_n) \in K[x_1, \ldots, x_n]$ such that $P(s_1, \ldots, s_2) = 0$ for all $s_1, \ldots, s_n \in S$.
For example, the sets $\{\sqrt{\pi}\}$ and $\{2\pi + 1\}$ are algebraically independent over $\mathbb{Q}$ because they are transcendental over $\mathbb{Q}$, but the set $\{\sqrt{\pi}, 2\pi+1\}$ is not. That is because $P(x, y) = 2x^2 - y + 1 \in \mathbb{Q}[x, y]$ has $P(\sqrt{\pi}, 2\pi + 1) = 0$.

Now back to the question, suppose there were a polynomial $P(x, y) \in \mathbb{C}[x, y]$ such that $P(E_4, E_6) = 0 \, \forall z \in \mathbb{H}$. Then I think the issue comes from the transformation law that upon $z \mapsto -\frac{1}{z}$:

$$E_4(z) \mapsto E_4(-\frac{1}{z}) = z^4 E_4(z),$$
$$E_6(z) \mapsto E_6(-\frac{1}{z}) = z^6 E_6(z).$$

Given that they don't transform the same way and $P(E_4, E_6) = 0 \, \forall z \in \mathbb{H}$, we must have that $P(x, y)$ is homogeneous. I mean it must be built from the monomials from part a.: $E_4^a E_6^b$ such that $4a + 6b = \deg(P) \in \mathbb{Z}_{\geq 0}$. Now I must somehow conclude that this is only possible when $P = 0$ altogether.

Closing comments

The last five problems felt significantly harder than the previous ten. I am still not satisfied with some, but for now I will continue with chapter three and hope to solve them later or ask help from someone more knowledgeable than I am. My solutions to 12a, 13b, 14c and 15 are incomplete.

3 Modular forms for congruence subgroups

Pensées

- I'm familiar with congruence in the 'classical geometry setting', but in what sense are these subgroups congruent (and to what)?
- If it is not in the problems: an example of a subgroup with finite index in $\mathrm{SL}_2(\mathbb{Z})$ that does not contain $\Gamma(N)$ for any N would be nice.
- I checked the claim at the bottom of p. 33 explicitly: indeed $\Gamma_1(2) = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\} = \Gamma_0(2)$.
- Failing to see what's going on in the explanation of the moduli interpretation of $\Gamma_0(N)$ on p.34... Ah, after some back and forth with ChatGPT[4] I see the punchline now. Namely, we are interested (for reasons I think come from elliptic curves, hopefully will be clear later) in inequivalent pairs (L, G) where L is a lattice and $G \subseteq \mathbb{C}/L$ a cyclic subgroup of order N and inequivalent means up to homothety. The text explains how G can be encoded into another lattice L', so studying $(L, G) \leftrightarrow$ studying (L, L'). Then, applying a modular transformation γ to L preserves the previous correspondence iff $\gamma \in \Gamma_0(N)$ so this in turn shows that studying homothety classes (L, G) is in bijection with the set $\Gamma_0(N) \setminus \mathbb{H}$.
- I should also comment that 'the inverse image of G in $\mathbb{C}$' means the following. We have the map $\pi : \mathbb{C} \longrightarrow \mathbb{C}/L \supseteq G$ so we define $L' = \{z \in \mathbb{C} |\, \pi(z) \in G\}$.
- The example on p.34 also needed clarification. $\Gamma \setminus \mathrm{SL}_2(\mathbb{Z})$ means $\Gamma g = \{\gamma g |\, \gamma \in \Gamma\}$ for $g \in \mathrm{SL}_2(\mathbb{Z})$ and g_1, g_2 represent the same coset if $\Gamma g_1 = \Gamma g_2 \iff g_1 g_2^{-1} \in \Gamma$. In this example $\Gamma = \Gamma_0(2)$ so for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ we must have the row $(c, d) \equiv (0, 1)$, $(1, 0)$ or $(1, 1)$ mod 2 (not $(0, 0)$ because $\det = 1$ in $\mathrm{SL}_2(\mathbb{Z})$). These three options precisely provide the three cosets as stated because for $g_{1/2} = \begin{pmatrix} a_{1/2} & b_{1/2} \\ c_{1/2} & d_{1/2} \end{pmatrix}$ we have $g_2^{-1} = \begin{pmatrix} d_2 & -b_2 \\ -c_2 & a_2 \end{pmatrix}$ so the lower left entry of $g_1 g_2^{-1} = c_1 d_2 - d_1 c_2$. Therefore, g_1, g_2 represent the same coset iff $c_1 d_2 - d_1 c_2 \equiv 0$ mod 2 so three matrices $\mathbf{1}, S, ST$ represent three different classes.
  The next part about the fundamental domain, the idea is that $\mathrm{SL}_2(\mathbb{Z}) = \Gamma \cup \Gamma S \cup \Gamma ST$ so we can always write $\mathrm{SL}_2(\mathbb{Z}) \ni g = \gamma r$, $\gamma \in \Gamma$ and $r \in \{\mathbf{1}, S, ST\}$ so $g\mathcal{D} = \gamma r \mathcal{D} \implies \mathcal{D}_\Gamma = \mathcal{D} \cup S(\mathcal{D}) \cup ST(\mathcal{D})$ as illustrated. This would not be possible for the right coset because then $g = r\gamma$ rahter than γr so we would not get an expression of the full $\mathrm{SL}_2(\mathbb{Z})$ as Γ translate.
- With the previous analysis, the content of proposition 3.2 becomes almost obvious as generalisation. Although, I do not fully understand the part about 'stable under -1' and $-1 \notin \Gamma$. It is confusing because $r, -r$ act the same on $\mathcal{D}$ so including both in R should not change $\mathcal{D}_\Gamma$...
- In the proof of lemma 3.3., why does it suffice to show that for every $t \in \mathbb{Q}$ there is a $\gamma \in G$ st $\gamma\infty = t$? Is it because $\mathrm{SL}_2(\mathbb{Z})$ is commonly known to act transitively on $\mathbb{Q}$? Ah no, it is because then $g_x\infty = x$ and $g_y\infty = y$ we can send $x \mapsto y$ by $g_y g_x^{-1} \in \mathrm{SL}_2(\mathbb{Z})$ and of course $\infty \mapsto \infty$ by the identity in $\mathrm{SL}_2(\mathbb{Z})$.

[4] I do allow myself to use AI to clarify the main text, not in solving problems.

- Why are cusps of Γ defined to be $\Gamma \setminus \mathbb{P}^1(\mathbb{Q})$? For example, if $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ we would find only ∞? This is true as we saw that $\mathrm{SL}_2(\mathbb{Z})$ acts transitively on $\mathbb{P}^1(\mathbb{Q})$ so we can represent the quotient by a single coset $\{\infty\}$. It remains to understand why this is a natural generalisation for cusps. A quick google search states the following definition in the Cambridge dictionary: A cusp is a noun that means a dividing line, transition point, or edge between two different states, or a sharp, pointed end where two curves meet. So I'm guessing this has to do with the relationship with elliptic curves and the cusps correspond to points where it would self intersect/be singular of some sort?

- This is the first time encountering $\twoheadrightarrow$ to denote surjective map and $\rightarrowtail$ for injective map haha. In this part though (p.35-36) it is obvious (canonical) how the surjection works (namely quotient map), but to me it is not clear how the injection works. Is it simply inclusion? In other words, are cosets $\Gamma_t \setminus (\gamma_t \mathrm{SL}_2(\mathbb{Z})_\infty \gamma_t^{-1})$ contained in the cosets $\Gamma \setminus \mathrm{SL}_2(\mathbb{Z})$?

- First I was confused why above lemma 3.4 we have all the equivalences showing $\gamma \in \Gamma_t \iff \gamma \in \gamma_t \mathrm{SL}_2(\mathbb{Z})_\infty \gamma_t^{-1}$ do we need to intersect with Γ in $\Gamma_t = \Gamma \cap \gamma_t \mathrm{SL}_2(\mathbb{Z})_\infty \gamma_t^{-1}$? But that is because $\mathrm{SL}_2(\mathbb{Z})_\infty$ is the stabiliser of ∞ for the full $\mathrm{SL}_2(\mathbb{Z})$ whereas $\gamma \in \Gamma$ by assumption at the top.

- Γ_t is of finite index in $\Gamma_t \setminus (\gamma_t \mathrm{SL}_2(\mathbb{Z})_\infty \gamma_t^{-1})$ because Γ is a congruent subgroup of $\mathrm{SL}_2(\mathbb{Z})$, so $H_{\mathfrak{c}}$ is of finite index in $\mathrm{SL}_2(\mathbb{Z})_\infty$. I see this is is poorly phrased, but I mean that the injective map defined above eq. 3.1 means that Γ_t is of finite index in $\gamma_t \mathrm{SL}_2(\mathbb{Z})_\infty \gamma_t^{-1}$ because Γ is in $\mathrm{SL}_2(\mathbb{Z})$ because it is a congruence subgroup.

- The definition of width and (ir)regularity of a cusp seems a bit strange. I mean it lacks intuition (to me at least) which again is probably found in its relation to elliptic curves.

- In lemma 3.5 part two a 'fibre' of a map is mentioned, which I only know in the context of bundles. I didn't know it's more general (Wikipedia): A fibre of a map $f : X \longrightarrow Y$ is an equivalence class in X for the relation of having the same image under f. If y is that image, then this fibre equals the inverse image $f^{-1}(y) = \{x \in X | f(x) = y\}$ of y and is called the fibre of f over y. So basically a fancy way of saying the pre-image of y...

- Quick comment on the last step of the proof of Lemma 3.5 part two: Assuming part 1 is well-known, we have $f : \mathrm{Stab}_H x \setminus \mathrm{Stab}_G x \longrightarrow H \setminus G$ with $\mathrm{im}(f) = H \setminus H\mathrm{Stab}_G x$ so using the isomorphism theorem the claim follows by injectivity.

- Corollary 3.6 confuses me... Is Cusps(Γ) a set of orbit representatives for the quotient $H \setminus X = \overline{\Gamma} \setminus \mathbb{P}^1(\mathbb{Q})$?

- The short discussion at 3.3 about the width or (ir)regular cusps I don't follow entirely, but think this will be clearer when attempting some problems about computing these widths. (So will postpone for now.)

- Not familiar with the complex analysis statement in the proof of thm 3.8: 'Since both sides...identically zero.'

- To be explicit for Cor.3.9:

$$\begin{aligned}\theta\Big(\frac{z}{4z+1}\Big) &= \theta\Big(-\frac{1}{4z'}\Big)\\ &= \sqrt{-2iz'}\theta(z') = \sqrt{-2iz'}\theta(z'+1)\\ &= \sqrt{-2i(-1/(4z)-1)}\theta\Big(-\frac{1}{4z}\Big)\\ &= \sqrt{2i\Big(\frac{4z+1)}{4z}\Big)}\sqrt{-2iz}\theta(z)\\ &= \sqrt{4z+1}\theta(z).\end{aligned}$$

- After a brief thought, I think to complete the argument of the proof of Thm. 3.10 the authors meant that $f \coloneqq \theta^2 = \Big(1+2\sum_{n=0}^{\infty} q^{n^2}\Big)^2 = 1+4\sum_{n=0}^{\infty} q^{n^2} + 4\Big(\sum_{n=0}^{\infty} q^{n^2}\Big)^2$ so the coefficients are all constants for each n; indeed there exist $C, d \in \mathbb{R}$ such that $|a_n| \leq Cn^d$ then. (?)

- As a first check of Thm. 3.11: it should generalise Thm.2.8 so suppose $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, then $\mathrm{Cusps}(\Gamma) = \{[\infty]\}$ and $\epsilon_{\Gamma,\mathfrak{c}} = 2$, because $-1 \in \Gamma$, and of course $(\mathrm{SL}_2(\mathbb{Z}) : \Gamma) = 1$. In order to match Thm. 2.8 we must therefore have that $\#\Gamma_i = 4$, $\#\Gamma_\rho = 6$ and $\#\Gamma_z = 2$ for all $z \in \mathcal{D}\backslash\{i, \rho\}$. This last bit is not immediately obvious to me, but I think the table at the bottom of p.12 can be used to verify it.
  Ah, indeed. Counting the number of matrices in $\mathrm{SL}_2(\mathbb{Z})$ that stabilise i, ρ in said table we find 4, 6 respectively which correspond to the orders of the stabilisers and we find 2 for the other z's.
  (And of course ρ, $\rho+1$ are in the same coset in $\mathrm{SL}_2(\mathbb{Z}) \setminus \mathbb{H}$. This, however, confuses me to think why $\mathcal{D}$ includes the right boundary and part of the arc... but I think this is convention where points on the boundary are identified.)

- Proof of Thm. 3.11 requires discussion.

- In paragraph 3.7 had to recall (from Wolfram Mathworld): The group torsion $\mathrm{Tor}(G)$ of a group G is the **set** of elements $g \in G$ such that $g^n = e$ for some natural number n, where e is the identity element of the group G. In the case that G is Abelian, $\mathrm{Tor}(G)$ is a subgroup and is called the torsion subgroup of G. If $\mathrm{Tor}(G)$ consists only of the identity element, the group G is called torsion-free.

  Reading this, it's clear the torision subgroup of $\mathbb{C}^*$ is the group of roots of unity. Also, the statement that since $(\mathbb{Z}/N\mathbb{Z})^*$ is finite means the image of χ' is contained in the torsion subgroup is clear too. Because finiteness of $(\mathbb{Z}/N\mathbb{Z})^*$ means that for each element there is an n such that $g^n = e$ which carries over to the image under χ' by the structure preservation of a group homomorphism.

- Pretty cool that Dirichlet characters modulo N form a group isomorphic to $(\mathbb{Z}/N\mathbb{Z})^*$ with order $\phi(N)$! (Do need to figure out the 'non-canonical' part...)

  As an aside, Euler's totient function:

$$\phi(N) = N\prod_{p|N}(1-\frac{1}{p})$$

- Very neat! Some foreshadowing: $\theta^k = \sum_{n=0}^{\infty} r_k(n)q^n$.

3.1

- $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \gamma \bmod N$ so writing $\gamma = \begin{pmatrix} \overline{a} & \overline{b} \\ \overline{c} & \overline{d} \end{pmatrix}$ we have $a = \overline{a} + \alpha N$ and $b = \overline{b} + \beta N$ for certain integers α, β. Also noting $\gamma \in \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ we must have $\overline{a}\overline{d} - \overline{b}\overline{c} = (a - \alpha N)\overline{d} - (b - \beta N)\overline{c} = a\overline{d} - b\overline{c} + (\beta - \alpha)N = 1$, but this is nothing other than the statement that $\gcd(a, b, N) = 1$ (as $\alpha, \beta, \overline{c}, \overline{d} \in \mathbb{Z}$).

- $\boxed{a = 0}$: By part a. we must have $\gcd(b, N) = 1$ in this case, so simply choose k such that $\gcd(k, b) = 1$, $l = 0$.

  $\boxed{a \neq 0}$: First I was confused about how we can always write $a = uv$ for u, v coprime in for example 64, but then realised we can always choose u or v to be 1 as well. We need to do that for prime powers, but otherwise we can write it as such for u, v not 1.
  I immediately realise $p|u \iff p \nmid v$, but I do not see how on earth there is a relation with b at this point. I know from the previous part that of course there are α, β, γ such that $\alpha a + \beta b + \gamma N = 1$, but don't see how that is of use...
  Assuming $p|u \iff p|b$, it is clear that a and $b + vN$ are coprime because $p \nmid v$ and p cannot divide N because modulo p, $\alpha a + \beta b + \gamma N = 1$ would read $0 \equiv \gamma N \equiv 1$ leading to a contradiction. This then shows that $\gcd(a + kN, b + lN) = 1$ for $k = 0$, $l = v$.

- For this final step I took inspiration from point 2 on this webpage: `https://surya-teja.com/2010/11/09/surjectivity-of-sl_nz-to-sl_n-z-nz/`. Namely, we consider the matrix

$$\begin{pmatrix} a & b' \\ c + mN & d + nN \end{pmatrix}, \; b' = b + vN$$

  with determinant $a(d + nN) - b'(c + mN) \equiv 1 \bmod N$ by construction, hence we can write $ad - b'c = 1 + tN$. Rewriting the determinant: $(ad - b'c) + N(an - b'm)$ we can choose n, m such that $an - b'm = -t$ because $\gcd(a, b') = 1$ by the previous bullet point. QED.[5]
  This then completes the construction of a matrix in $\mathrm{SL}_2(\mathbb{Z})$ that reduces to γ modulo N showing surjectivity. We now immediately see indeed that $\mathrm{SL}_2(\mathbb{Z})/\Gamma(N) \simeq \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$.

3.2

My immediate thought is that proving normality of $\Gamma_1(N)$ in $\Gamma_0(N)$ follows directly from $\Gamma_0(N)/\Gamma_1(N) \simeq \left(\mathbb{Z}/\mathbb{N}\mathbb{Z}\right)^*$ as the kernel of a group homomorphism is a normal subgroup. So we only need to show $\ker(\phi) = \Gamma_1(N)$ for

$$\begin{aligned} \phi : \Gamma_0(N) &\longrightarrow \left(\mathbb{Z}/\mathbb{N}\mathbb{Z}\right)^* \\ \begin{pmatrix} a & b \\ c & d \end{pmatrix} &\mapsto d \bmod N. \end{aligned}$$

Group homomorphism:

- Well-defined, as $d \not\equiv 0 \bmod N$ for any $\gamma \in \Gamma_0(N)$ by unity of the determinant, ✓
- $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mapsto 1 \bmod N$, ✓

[5] The important clue I initially missed is that of course $\gcd(x, y) = 1 \iff \exists \alpha, \beta : \; \alpha x + \beta y = 1$ also means $t\alpha x + t\beta y = t$.

- $\phi\Big(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}\Big) = \phi\Big(\begin{pmatrix} \cdots & \cdots \\ \cdots & b'c+dd' \end{pmatrix}\Big) = dd' \bmod N$ (because $c \equiv 0 \bmod N$) so this equals $\phi\Big(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\Big)\phi\Big(\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}\Big)$, ✓

And it is clear that this group homomorphism is surjective so we must check whether $\ker(\phi) = \Gamma_1(N)$ now. Well, clearly $\Gamma_1(N) \subseteq \ker(\phi)$ and conversely we observe that for any $\gamma \in \ker(\phi)$ it must have d'th element being $1 + tN$ for some $t \in \mathbb{Z}$. Also, as $\gamma \in \Gamma_0(N)$ the c'th element must be a multiple of N: $c = xN$ for some $x \in \mathbb{Z}$. Hence $\det\gamma = a(1+tN) - bxN \overset{!}{=} 1 \iff a = \frac{1+bxN}{1+tN} \in \mathbb{Z}$ meaning $t = 0$ or $bx = t$; both cases $a \equiv 1 \bmod N$ so indeed $\gamma \in \Gamma_1(N)$.
Lastly, by the isomorphism theorem and surjectivity of ϕ, we conclude $\Gamma_0(N)/\Gamma_1(N) \simeq \big(\mathbb{Z}/\mathbb{N}\mathbb{Z}\big)^*$.

3.3

(I think this question is either a bit trivial or I'm missing something haha, in any case here is what I think:)

a. Given that $\Gamma(DN) \subseteq \Gamma(N)$ we can simply compute $\beta\gamma\beta^{-1}$ ($\beta \in \mathrm{GL}_2^+(\mathbb{Z})$) and see if it is in $\Gamma(DN)$ for

$$\beta = \begin{pmatrix} a & b \\ c & d \end{pmatrix},\ \det(\beta) = D,\ \gamma \in \Gamma(DN) \implies$$
$$\beta\gamma\beta^{-1} \equiv \frac{1}{D}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \bmod DN$$

showing $\beta\Gamma(DN)\beta^{-1} \subseteq \Gamma(N)$.

b. Well, $\beta\Gamma(DN)\beta^{-1} \subseteq \Gamma(N) \iff \Gamma(DN) \subseteq \Gamma(N) \cap \beta^{-1}\Gamma(N)\beta$ (by noting that $\Gamma(DN) \cap \Gamma(N) = \Gamma(DN)$).

c. We did not use that the entries need to be in $\mathbb{Z}$ anywhere so I think we still have $\Gamma(DN) \subseteq \Gamma(N) \cap \alpha^{-1}\Gamma(N)\alpha = \Gamma'$ for $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ ($+$ indicating $\det(\alpha) = D > 0$). This immediately demonstrates Γ' is a congruence subgroup with level DN.

3.4

So it has been proved that for any $z \in \mathbb{H}\ \exists\gamma \in \Gamma$ such that $\gamma z \in \mathcal{D}_\Gamma$. Now for uniqueness, suppose that $\gamma z,\ \tilde{\gamma} z \in \mathcal{D}_\Gamma$ for $\gamma \neq \tilde{\gamma}$. Then,

$$\gamma z = z_0,\ \tilde{\gamma} z = \tilde{z}_0 \implies$$
$$z = \gamma^{-1} z_0 = \tilde{\gamma}^{-1}\tilde{z}_0 \iff$$
$$\tilde{\gamma}\gamma^{-1} z_0 = \tilde{z}_0$$

but this means that $z_0,\ \tilde{z}_0$ are in the same orbit which is not possible by construction in the interior of $\mathcal{D}_\Gamma$. For the interior we must therefore have $z_0 = \tilde{z}_0$ in which case too we have $\gamma\tilde{\gamma}^{-1} z = z \iff$ z is fixed under $\gamma\tilde{\gamma}^{-1}$, but z is arbitrary so we must have $\tilde{\gamma} = \gamma$ (for the interior points of $\mathcal{D}_\Gamma$). The action of $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ is independent of multiplication by $\{\pm 1\}$ so the conclusion follows.

3.5

Is this the same as showing all $H_{\mathfrak{c}}$ are isomorphic for every $t \in \mathbb{P}^1(\mathbb{Q})$ or is it stronger?
I think they mean stronger; I think they are asking to show

$$\begin{aligned} H_{\mathfrak{c}} = \gamma_t^{-1}\Gamma\gamma_t \cap \mathrm{SL}_2(\mathbb{Z})_\infty &= \gamma_{t'}^{-1}\Gamma\gamma_{t'} \cap \mathrm{SL}_2(\mathbb{Z})_\infty = H'_{\mathfrak{c}} \\ &\Longleftarrow \gamma_t^{-1}\Gamma\gamma_t = \gamma_{t'}^{-1}\Gamma\gamma_{t'} \text{ (this might be iff, but won't write that to be sure)} \\ \Longleftrightarrow\ (\gamma_{t'}\gamma_t^{-1})\Gamma(\gamma_t\gamma_{t'}^{-1}) &= \Gamma. \end{aligned}$$

Writing $h \coloneqq \gamma_{t'}\gamma_t^{-1}$, it boils down to showing $h\Gamma h^{-1}$ indeed equals Γ. Should we then perhaps show $h \in \Gamma$? Because then it is immediate.